

1 BRST in Abstraction

Consider a generic quantum field theory, with fields denoted by Φ_i . The index i is an abstract notation, which also includes the spacetime/worldsheet variable as well. For example, in string theory, the fields will be $X^\mu(\sigma)$ and $g_{ab}(\sigma)$. Let the field theory be gauge invariant under a gauge transformation,

$$\Phi_i \rightarrow \Phi_i + \delta_g \Phi_i \quad (1.1)$$

We can define a basis for gauge transformations, $\{\delta_\alpha\}$ so that a general gauge transformation can be written as,

$$\delta_g = \epsilon^\alpha \delta_\alpha \quad (1.2)$$

This will be clear when we consider examples in the upcoming sections. Let the gauge transformations obey a closed algebra,

$$[\delta_\alpha, \delta_\beta] = f_{\alpha\beta}^\gamma \delta_\gamma \quad (1.3)$$

The gauge redundancy is fixed by a gauge fixing condition,

$$F^A(\Phi) = 0 \quad (1.4)$$

All the above indices are abstract, as mentioned before. The path integral quantization of the theory is carried out by calculating the partition function,

$$Z = \frac{1}{\text{Vol}} \int \mathcal{D}\Phi e^{-S_0} \quad (1.5)$$

Where “Vol” indicates the fact that the path integral has a huge redundancy due to gauge symmetry. One must fix the gauge in order to get sensible results out of this. Now, in the Faddeev-Popov procedure, the gauge is fixed by inserting 1 into the path integral in the form,

$$1 = \int \mathcal{D}\epsilon \delta(F^A(\Phi)) \det\{\delta_\alpha F^A(\Phi)\} \quad (1.6)$$

The functional integral is over all the gauge transformations. The delta function can be re-written in terms of another functional integral as,

$$\delta(F^A(\Phi)) = \int \mathcal{D}B_A \exp\{iB_A F^A(\Phi)\} \quad (1.7)$$

The Faddeev-Popov determinant can be written using functional integrals of Grassmann odd fields,

$$\det\{\delta_\alpha F^A(\Phi)\} = \int \mathcal{D}b_A \mathcal{D}c^\alpha \exp\{-b_A c^\alpha \delta_\alpha F^A(\Phi)\} \quad (1.8)$$

Now, the functional integral over gauge transformations can be performed, which cancels the “Vol” factor. We will be left with,

$$Z = \int \mathcal{D}\Phi \mathcal{D}B_A \mathcal{D}b_A \mathcal{D}c^\alpha \exp\{-S\} \quad (1.9)$$

where,

$$S = S_0 - iB_A F^A(\Phi) + b_A c^\alpha \delta_\alpha F^A(\Phi) \quad (1.10)$$

The gauge symmetry has been fixed, at the expense of new fields in the game.

BRST

BRST symmetry is a global symmetry of the gauge fixed action (1.10). To understand this, let us promote the gauge parameter ϵ^α to the ghost field c^α . That is, we now *demand* that,

$$\delta_B \Phi_i = -i\epsilon c^\alpha \delta_\alpha \Phi_i \quad (1.11)$$

is a symmetry of (1.10). Notice that the above is just a gauge transformation with the gauge parameter replaced by the ghost field c . The BRST parameter is now ϵ in the above equation, not c . Since ϵ is not coordinate dependent (it has no index), the symmetry is a global one. Also, ϵ must be a Grassmann odd variable for consistency. We now determine how the other fields must transform in order for (1.11) to be a symmetry of the gauge fixed action. We know that S_0 is already invariant under this transformation. This is because S_0 was originally gauge invariant, and (1.11) is nothing but a gauge transformation with the parameter ϵc^α . Therefore, we should demand,

$$\delta_B (-iB_A F^A(\Phi) + b_A c^\alpha \delta_\alpha F^A(\Phi)) = 0 \quad (1.12)$$

This gives,

$$-i(\delta_B B_A) F^A(\Phi) - iB_A (\delta_B F^A(\Phi)) + (\delta_B b_A) c^\alpha \delta_\alpha F^A(\Phi) + b_A (\delta_B c^\alpha) \delta_\alpha F^A(\Phi) + b_A c^\alpha \delta_B \delta_\alpha F^A(\Phi) = 0 \quad (1.13)$$

From (1.11), we can infer,

$$\delta_B F^A(\Phi) = -i\epsilon c^\alpha \delta_\alpha F^A(\Phi) \quad (1.14)$$

Plugging this in, we have,

$$-i(\delta_B B_A) F^A(\Phi) + (-\epsilon B_A + \delta_B b_A) c^\alpha \delta_\alpha F^A(\Phi) + b_A (\delta_B c^\alpha \delta_\alpha F^A + c^\alpha c^\beta (-i\epsilon) \delta_\beta \delta_\alpha F^A) = 0 \quad (1.15)$$

In the third term, due to the anti commuting nature of c 's, we can replace,

$$\delta_\alpha \delta_\beta \rightarrow \frac{1}{2} [\delta_\alpha, \delta_\beta] = \frac{1}{2} f_{\alpha\beta}^\gamma \delta_\gamma \quad (1.16)$$

Thus we have,

$$-i(\delta_B B_A)F^A(\Phi) + (-\epsilon B_A + \delta_B b_A)c^\alpha \delta_\alpha F^A(\Phi) + b_A \left(\delta_B c^\gamma - \frac{i}{2} \epsilon c^\alpha c^\beta f_{\alpha\beta}^\gamma \right) = 0 \quad (1.17)$$

Staring at this for a while, we can infer the full set of BRST transformations,

$$\begin{aligned} \delta_B \Phi_i &= -i\epsilon c^\alpha \delta_\alpha \Phi_i \\ \delta_B B_A &= 0 \\ \delta_B b_A &= \epsilon B_A \\ \delta_B c^\alpha &= \frac{i}{2} \epsilon f_{\beta\gamma}^\alpha c^\beta c^\gamma \end{aligned} \quad (1.18)$$

Here are some remarks:

- We can assign ghost numbers to all the fields. Say c gets a number 1, then for consistency ϵ must have -1 . We expect B_A to be of ghost number zero, therefore b also should have -1 .
- We can construct the ghost+gauge fixing part of the action itself as a BRST variation.

$$\delta_B(b_A F^A) = \epsilon B_A F^A + b_A(-i\epsilon c^\alpha \delta_\alpha F^A)$$

This gives us a way to justify the Faddeev-Popov procedure. We start with S_0 only, and demand that BRST is the actual symmetry of the problem. Then adding the above term is consistent with this symmetry because $\delta_B \circ \delta_B = 0$ (this will be shown shortly).

- Since BRST is a global symmetry, there is a conserved charge, denoted by Q_B . We can then write the BRST transformations as (anti)-commutations relations with the BRST charge. For a general operator $\mathcal{O}$, the change under BRST is,

$$\delta_B \mathcal{O} = -i[\mathcal{O}, \epsilon Q_B]_- = -i\epsilon[\mathcal{O}, Q_B]_\pm$$

In the third equality, the commutator changed to either commutator or anti commutator depending upon $\mathcal{O}$ being Grassmann odd or even. This is because ϵ is Grassmann odd.

Necessary Condition for Physical States

Consider a small change in the gauge fixing condition, δF^A . This changes the ghost + gauge fixing part of the action by,

$$\delta_B(b_A \delta F^A) = -i\epsilon \{b_A \delta F^A, Q_B\} \quad (1.19)$$

If we write the inner-product between any two states as a path integral,

$$\langle f|i\rangle = \int_i^f \mathcal{D}(\text{fields}) e^{-S} \quad (1.20)$$

then this change in gauge fixing function will change the inner-product by,

$$\delta \langle f|i\rangle = \int_i^f \mathcal{D}(\text{fields}) e^{-S} \underbrace{\delta S}_{=\delta_B(b_A \delta F^A)} = \langle f| \delta_B(b_A \delta F^A) |i\rangle = -i\epsilon \langle f| \{b_A \delta F^A, Q_B\} |i\rangle \quad (1.21)$$

For physical states, the choice of gauge should not matter, hence change in the gauge fixing function should not be a problem. Hence for physical states, we must have,

$$\langle \Psi_f | \{b_A \delta F^A, Q_B\} | \Psi_i \rangle = 0 \quad (1.22)$$

For this to be true for any arbitrary δF^A , we must have,

$$\hat{Q}_B |\Psi\rangle = 0 \quad \text{for } |\Psi\rangle \text{ being a physical state} \quad (1.23)$$

The necessary condition of the physical states is that they must be BRST invariant.

An Aside : A Family of Gauges

To the action, we can add another BRST invariant term,

$$\frac{1}{\epsilon} \delta_B(b_A B_C M^{AC}) = -B_A B_C M^{AC} \quad (1.24)$$

for a constant matrix M . This new term does not change the argument above of matrix elements between physical states being gauge choice independent. With this new term, the integral over B is now Gaussian, and will produce a family of gauges.

Nilpotency of δ_B

The BRST charge must be conserved. This ensures that if we move along the space of gauge choices, we will still have the property of physical states being unaffected by the choice of gauge. Let $\hat{H}$ be the Hamiltonian. Then the BRST charge being conserved just translates to,

$$[Q_B, H] = 0 \quad (1.25)$$

Now, imagine we make a change to the Hamiltonian by changing the choice of gauge. We already saw that this change is proportional to,

$$\delta H \propto \{Q_B, b_A \delta F^A\} \quad (1.26)$$

For the BRST charge to be conserved with every change in the Hamiltonian, we must have,

$$\begin{aligned} 0 &= [Q_B, \{Q_B, b_A \delta F^A\}] \\ &= Q_B^2 b_A \delta F^A + Q_B b_A \delta F^A Q_B - Q_B b_A \delta F^A Q_B - b_A \delta F^A Q_B^2 \\ &= [Q_B^2, b_A \delta F^A] \end{aligned} \quad (1.27)$$

For this to be true for all δF^A , we must have,

$$Q_B^2 = \text{constant} \quad (1.28)$$

It turns out that this constant must be zero. One way to see it is to look at the ghost number of Q_B . The BRST change in any operator is,

$$\delta_B \mathcal{O} = -i\epsilon [\mathcal{O}, Q_B]_{\pm} \quad (1.29)$$

Since ϵ has a ghost number -1 , Q_B must have a ghost number $+1$ for consistency. Therefore, Q_B^2 has ghost number 2. Hence the constant should have a ghost number 2. This tells us that the constant must be zero.

$$Q_B^2 = 0 \quad (1.30)$$

Not completely clear about how to explain this better.

BRST Cohomology and Physical States

Since the BRST charge is nilpotent, it has some interesting consequences¹. The nilpotency condition (4.60) is an operator equation, meaning $Q_B^2 |\Psi\rangle = 0$, $\forall |\Psi\rangle \in \mathcal{F}$, the Fock space².

¹One of them is that the eigen-values of Q_B have to be Grassmann-valued.

²this is the entire Fock space, not the physical set of states

Let us define a couple of subsets of $\mathcal{F}$,

$$\text{Im}\{Q_B\} := \left\{ |\Psi\rangle \in \mathcal{F} \mid \exists |\Phi\rangle \in \mathcal{F} : |\Psi\rangle = Q_B |\Phi\rangle \right\} \quad (1.31)$$

In plain English, what this says is that any vector in the *image* of Q_B can be written as Q_B acting on some other vector. We can also define,

$$\ker\{Q_B\} := \left\{ |\Psi\rangle \in \mathcal{F} \mid Q_B |\Psi\rangle = 0 \right\} \quad (1.32)$$

These are vectors which are annihilated by Q_B , called the *kernal* of Q_B . Since $Q_B^2 = 0$, we have an additional relation,

$$\text{Im}\{Q_B\} \subset \ker\{Q_B\} \quad (1.33)$$

We notice that,

$$|\Psi\rangle \in \text{Im}\{Q_B\} \Rightarrow \langle \Psi | \Psi \rangle = 0 \quad (1.34)$$

Therefore, there are non zero vectors in the image which are not normalizable. We want all our physical states to be normalizable. Clearly, we cannot choose the entire Fock space $\mathcal{F}$ to be our physical set of states. Looks like $\ker\{Q_B\}$ is a good candidate for $\mathcal{H}$. States in the kernal will trivially satisfy the nilpotency condition for Q_B . But since $\text{Im}\{Q_B\} \subset \ker\{Q_B\}$, and vectors in $\text{Im}\{Q_B\}$ have zero norm, we cannot choose the physical Hilbert space to be $\ker\{Q_B\}$ also. One possibility is to just take away the image from the kernal. This will result in taking away the zero element from the kernal, then we cannot make the kernal into a Hilbert space. We also notice that if $|\Psi\rangle \in \ker\{Q_B\}$ and has a non zero norm, and $|\Phi\rangle \in \text{Im}\{Q_B\}$, then $|\Psi\rangle + |\Phi\rangle \in \ker\{Q_B\}$ and $|\Psi\rangle + |\Phi\rangle$ is normalizable. That is, we can add a vector from the image to the physical state and nothing changes. This gives us an idea: first, define an equivalence relation,

$$|\Psi\rangle \sim |\Phi\rangle \quad \text{if} \quad |\Psi\rangle - |\Phi\rangle \in \text{Im}\{Q_B\} \quad (1.35)$$

Then we define a new set called the *Cohomology* of Q_B as,

$$\text{Coh}\{Q_B\} := \ker\{Q_B\} / \text{Im}\{Q_B\} \quad (1.36)$$

The cohomology is the *Quotient* of the kernal with respect to the equivalence class. What this means is that, if two elements in the kernal only differ by a vector from the image, then these two elements in the kernal are identified and is one element in the cohomology. This way, all vectors in $\text{Im}\{Q_B\}$ are the same element in the cohomology, namely the zero element. Hence, we define the physical set of states,

$$\mathcal{H} = \text{Coh}\{Q_B\} \quad (1.37)$$

Let us verify that once we start with $|\Psi\rangle \in \text{Coh}\{Q_B\}$, we stay in $\text{Coh}\{Q_B\}$ as time progress. This is easy. If $Q_B |\Psi\rangle = 0$, and $\langle\Psi|\Psi\rangle = 1$ say, then,

$$|\Psi(t)\rangle = e^{-iHt} |\Psi(0)\rangle \quad (1.38)$$

Then since $[Q_B, H] = 0$,

$$Q_B |\Psi(t)\rangle = Q_B e^{-iHt} |\Psi(0)\rangle = e^{-iHt} Q_B |\Psi(0)\rangle = 0 \quad (1.39)$$

Therefore, time evolution keeps the state in the cohomology.

2 The Point Particle

Let us now focus on a concrete example, the example of a point particle.

The Action

We know that the point particle is described by an action,

$$S = -m \int d\tau \sqrt{-\dot{x}^2} \quad (2.1)$$

The “dot” here is a derivative with respect to τ . The action has a weird square root, and it does not work for $m = 0$. It is convenient to re-write the action in terms of an auxiliary field, called the *einbein*,

$$S = \int d\tau \frac{1}{2} (e^{-1} \dot{x}^2 - e m^2) \quad (2.2)$$

We can easily convince ourselves that finding the EOM for e and substituting it back in the action gives us back (2.1). This new action, which is called the *Polyakov action*, does not have a square root, and it works for massless cases as well!

Reparameterization Invariance

The original action (2.1) had a symmetry of reparameterization. The Polyakov action also should have the symmetry. If we make a reparameterization,

$$\tau \rightarrow \tilde{\tau} = \tilde{\tau}(\tau) \quad (2.3)$$

Then,

$$\tilde{S} = \int \dot{\tilde{\tau}} d\tau \frac{1}{2} \left(\tilde{e}^{-1} \left(\frac{\dot{x}^2}{\dot{\tilde{\tau}}^2} \right) - \tilde{e} m^2 \right) \quad (2.4)$$

We infer that for $\tilde{S} = S$, we must have,

$$\dot{\tilde{\tau}} \tilde{e} = e \quad (2.5)$$

Let us consider an infinitesimal re-parameterization,

$$\tau \rightarrow \tau - \epsilon \xi(\tau) \quad (2.6)$$

Then,

$$\delta x(\tau) = -\epsilon \xi(\tau) \dot{x}(\tau) \quad (2.7)$$

$$\tilde{e} = \frac{e}{1 - \epsilon \dot{\xi}} = e(1 + \epsilon \dot{\xi}) \quad (2.8)$$

or,

$$\delta e(\tau) = \epsilon e(\tau) \dot{\xi}(\tau) \quad (2.9)$$

In summary, the following transformations keep the action intact,

$$\begin{aligned} \delta \tau &= -\epsilon \xi(\tau) \\ \delta x(\tau) &= -\epsilon \xi(\tau) \dot{x}(\tau) \\ \delta e(\tau) &= \epsilon e(\tau) \dot{\xi}(\tau) \end{aligned} \quad (2.10)$$

Faddeev-Popov Quantization

We try to quantize the action (2.2) using the path integral method. This boils down to calculating the partition function,

$$Z = \int \mathcal{D}x \mathcal{D}e \exp \left\{ - \int d\tau \frac{1}{2} (e^{-1} \dot{x}^2 + e m^2) \right\} \quad (2.11)$$

Performing the path integral over e seems to be impossible because of the e^{-1} in the exponent. But because of the re-parameterization symmetry, we can set the value of e to be a constant, 1 for instance. Let's see how this can be done. Say we have an e that is an arbitrary function of τ to start with. Then, we make a gauge transformation with the parameter ξ such that in the new gauge, we have,

$$\tilde{e}(\tau) = 1 = e(1 + \epsilon \dot{\xi}(\tau)) \quad (2.12)$$

The choice of ξ is then,

$$\epsilon \xi(\tau) = \int d\tau (e^{-1}(\tau) - 1) + \text{constant} \quad (2.13)$$

There are a class of choices for ξ , all of them differing by a real constant. [is this some kind of “Gribov ambiguity”? How is it that we take the gauge slicing picks out one choice of \$\xi\$ when we do Faddev-Popov?](#)

We then gauge fix by inserting a delta functional of ξ which chooses a ξ_0 such that $e = 1$.

$$1 = \int \mathcal{D}\xi \delta(\xi(\tau) - \xi_0(\tau)) = \int \mathcal{D}\xi \det \left(\frac{\delta e}{\delta \xi} \right) \delta(e(\tau) - 1) \quad (2.14)$$

We can also write,

$$\delta(e - 1) = \int \mathcal{D}B \exp \left\{ i \int d\tau B(\tau) (e(\tau) - 1) \right\} \quad (2.15)$$

Where B is an auxiliary field/Lagrange multiplier. Also,

$$\frac{\delta e(\tau)}{\delta \xi(\tau')} = e(\tau) \frac{d}{d\tau} \delta(\tau - \tau') \quad (2.16)$$

Hence,

$$\det \left(\frac{\delta e}{\delta \xi} \right) = \int \mathcal{D}b \mathcal{D}c \exp \left\{ - \int d\tau e b \dot{c} \right\} \quad (2.17)$$

Where b and c are *Faddeev-Popov* ghosts. Therefore the gauge fixed action becomes,

$$S = \int d\tau \left(\frac{1}{2} e^{-1} \dot{x}^2 + \frac{1}{2} e m^2 - iB(e - 1) + e b \dot{c} \right) \quad (2.18)$$

BRST

Let us now write down the BRST transformations (1.18) for the point particle. First, we need a basis for the gauge transformations, which are the reparameterization invariance. We can write,

$$\delta \tau = -\epsilon \xi(\tau) = \int d\tau' (-\epsilon \xi(\tau')) \delta(\tau - \tau') \quad (2.19)$$

Comparing with (1.2), we can infer,

$$\delta_{\tau_1} \tau = -\delta(\tau - \tau_1) \quad (2.20)$$

By the same logic,

$$\delta x^\mu(\tau) = -\epsilon \int d\tau_1 \xi(\tau_1) \delta(\tau - \tau_1) \dot{x}^\mu(\tau) \quad (2.21)$$

which tells us,

$$\delta_{\tau_1} x^\mu(\tau) = -\delta(\tau - \tau_1) \partial_\tau x^\mu(\tau) \quad (2.22)$$

and,

$$\delta e(\tau) = \int d\tau_1 (-\epsilon \xi(\tau_1)) \partial_\tau [\delta(\tau - \tau_1) e(\tau)] \quad (2.23)$$

tells us,

$$\delta_{\tau_1} e(\tau) = -\partial_\tau [\delta(\tau - \tau_1) e(\tau)] \quad (2.24)$$

The algebra of gauge transformations is then easy to find,

$$\begin{aligned} [\delta_{\tau_1}, \delta_{\tau_2}] x^\mu(\tau) &= -[\delta(\tau - \tau_1) \partial_\tau \delta(\tau - \tau_2) - \delta(\tau - \tau_2) \partial_\tau \delta(\tau - \tau_1)] \partial_\tau x^\mu(\tau) \\ &= \int d\tau_3 [\delta(\tau_3 - \tau_1) \partial_{\tau_3} \delta(\tau_3 - \tau_2) - \delta(\tau_3 - \tau_2) \partial_{\tau_3} \delta(\tau_3 - \tau_1)] (-\delta(\tau - \tau_3) \partial_\tau x^\mu(\tau)) \\ &= \int d\tau_3 f_{\tau_1 \tau_2}^{\tau_3} \delta_{\tau_3} x^\mu(\tau) \end{aligned} \quad (2.25)$$

where the structure functions are,

$$f_{\tau_1\tau_2}^{\tau_3} = \delta(\tau_3 - \tau_1)\partial_{\tau_3}\delta(\tau_3 - \tau_2) - \delta(\tau_3 - \tau_2)\partial_{\tau_3}\delta(\tau_3 - \tau_1) \quad (2.26)$$

Now, we are ready to write down the BRST transformations (1.18). For the position,

$$\delta_B x^\mu(\tau) = i\epsilon c(\tau)\dot{x}^\mu(\tau) \quad (2.27)$$

and for the einbein,

$$\delta_B e(\tau) = -i\epsilon\dot{c}(\tau)e(\tau) \quad (2.28)$$

This differs from Polchinski 4.2.22b. I do not know the discrepancy. But after the EOM has been plugged in, both matches.

As well as,

$$\delta_B B(\tau) = 0 \quad (2.29)$$

which is always the case and,

$$\delta_B b(\tau) = \epsilon B(\tau) \quad (2.30)$$

To find the BRST transformation of the ghost field c , we need to use the algebra of gauge transformations.

$$\begin{aligned} \delta_B c(\tau) &= \frac{i}{2}\epsilon \int d\tau_1 d\tau_2 f_{\tau_1\tau_2}^\tau c(\tau_1)c(\tau_2) \\ &= i\epsilon c(\tau)\dot{c}(\tau) \end{aligned} \quad (2.31)$$

It is convenient to integrate out B and thereby setting $e = 1$. Let us do that. Then the action (2.18) becomes,

$$S = \int d\tau \left(\frac{1}{2}\dot{x}^2 + \frac{1}{2}m^2 + b\dot{c} \right) \quad (2.32)$$

The BRST transformations now reduces to,

$$\begin{aligned} \delta_B x^\mu(\tau) &= i\epsilon c(\tau)\dot{x}^\mu(\tau) \\ \delta_B b(\tau) &= -i\epsilon \left(-\frac{1}{2}\dot{x}^2 + \frac{1}{2}m^2 + b\dot{c} \right) \\ \delta_B c(\tau) &= i\epsilon c(\tau)\dot{c}(\tau) \end{aligned} \quad (2.33)$$

where we plugged in the value of B by finding the EOM for e . We can use the EOM for c , namely $\dot{c} = 0$ also in the above transformations. The nilpotency condition now holds only up to EOM.

BRST Charge Q_B

Now, let us find the BRST charge via the Noether's theorem. First, let us integrate by parts the bc term in the action,

$$S = \int d\tau \left(\frac{1}{2} \dot{x}^2 + \frac{1}{2} m^2 - \dot{b}c \right) \quad (2.34)$$

This integration by parts seems to be crucial. If I don't do this, then I am not getting the correct BRST charge. I don't know what is going wrong!

First, let us find the canonical conjugate variables. In Euclidean signature, we have,

$$p^\mu = \frac{1}{-i} \frac{\partial L}{\partial \dot{x}_\mu} = i\dot{x}^\mu \quad (2.35)$$

The factor of i in front is because $t = -i\tau$.

Now we can find the BRST variation of the action (which should be zero),

$$\delta_B S = \int d\tau \left(\dot{x}^\mu \delta_B \dot{x}_\mu + 0 - (\delta_B \dot{b})c - \dot{b} \delta_B c \right) \quad (2.36)$$

Since $\delta_B c = 0$ when EOM is obeyed, we can ignore this term. We therefore have,

$$0 = \int d\tau \left(\frac{d}{d\tau} (\dot{x}^\mu \delta_B x_\mu) - \frac{d}{d\tau} ((\delta_B b)c) \right) \quad (2.37)$$

Therefore the Noether charge is,

$$Q_B = \dot{x}^\mu i\epsilon c \dot{x}_\mu - (-i\epsilon) \left(-\frac{1}{2} \dot{x}^2 + \frac{1}{2} m^2 + b\dot{c} \right) c \quad (2.38)$$

The $b\dot{c}$ term can be discarded because it is zero when we apply the EOM. This gives,

$$Q_B = \frac{1}{2} c(-p^2 + m^2) \quad (2.39)$$

Something went wrong!

Let me explain another way to find the BRST charge, mostly by inspection. We use the BRST transformations along with the canonical (anti) commutation relations to figure out Q_B . First, we have,

$$\delta_B x^\mu(\tau) = -i\epsilon [x^\mu(\tau), Q_B] = i\epsilon c(\tau) \dot{x}^\mu(\tau) \quad (2.40)$$

This tells us that Q_B must have a piece like,

$$Q_B = \frac{1}{2} c p^2 + (\text{stuff that commutes with } x^\mu) \quad (2.41)$$

We can check,

$$-i\epsilon \left[x^\mu, \frac{1}{2}cp^2 \right] = -i\epsilon \frac{1}{2}c[x^\mu, p^2] = (-i\epsilon)icp = i\epsilon c\dot{x} \quad (2.42)$$

where we used $p^\mu = i\dot{x}^\mu$. Let us use the BRST for b next. We have,

$$\delta_B b = -i\epsilon \{b, Q_B\} = -i\epsilon \left(-\frac{1}{2}\dot{x}^2 + \frac{1}{2}m^2 + b\dot{c} \right) = -i\epsilon \left(\frac{1}{2}p^2 + \frac{1}{2}m^2 \right) \quad (2.43)$$

We discarded the $\dot{c}$ term using the EOM. Also,

$$\{b, c\} = 1 \quad (2.44)$$

Hence, we have to have,

$$Q_B = \frac{1}{2}c(p^2 + m^2) \quad (2.45)$$

Let's check,

$$-i\epsilon \{b, Q_B\} = -i\epsilon \frac{1}{2}(p^2 + m^2) \quad (2.46)$$

Now, this structure of Q_B is already consistent with the c transformation, because we want $\{c, Q_B\} = 0$ using the EOM. Hence,

$$Q_B = \frac{1}{2}c(p^2 + m^2) = cH \quad (2.47)$$

where H is the Hamiltonian. The Hamiltonian acting on physical states must be zero. This is a constraint, which originally was due to the EOM of e . But now, we have fixed $e = 1$ by introducing other fields into the system. But the constraint equation manifests as the BRST invariance. This turns out to be general. The BRST operator takes the form of ghost $\times$ constraint.

Spectrum

We have the canonical (anti) commutation relations,

$$[x^\mu, p^\nu] = i\eta^{\mu\nu} \quad \text{and} \quad \{b, c\} = 1 \quad (2.48)$$

The second of the above requires some explanation. The canonical momentum conjugate to b is,

$$\pi_b = \frac{1}{-i} \frac{\partial \mathcal{L}}{\partial \dot{b}} \quad (2.49)$$

The $-i$ is due to the Euclidean signature, $t = -i\tau$, and the derivative is a “right-derivative”. If we apply this right derivative to (2.34), we get,

$$\pi_b = ic \quad (2.50)$$

Therefore, the canonical anti commutation relations give,

$$\{b, \pi_b\} = i \implies \{b, c\} = 1 \quad (2.51)$$

The bc algebra looks like that of a creation-annihilation algebra of a fermionic harmonic oscillator. This makes two states. Therefore, a complete set of states can be thought of as eigen states of the momentum operator, and the states generated by bc algebra. Let us denote these states by $|k, \uparrow\rangle$ and $|k, \downarrow\rangle$. We have,

$$\hat{p}^\mu |k, \text{any}\rangle = k^\mu |k, \text{any}\rangle \quad (2.52)$$

$$\hat{b} |k, \downarrow\rangle = 0 \quad \text{and} \quad \hat{b} |k, \uparrow\rangle = |k, \downarrow\rangle \quad (2.53)$$

$$\hat{c} |k, \downarrow\rangle = |k, \uparrow\rangle \quad \text{and} \quad \hat{c} |k, \uparrow\rangle = 0 \quad (2.54)$$

The BRST charge acts on these states to give,

$$\hat{Q}_B |k, \uparrow\rangle = 0 \quad \text{and} \quad \hat{Q}_B |k, \downarrow\rangle = \frac{1}{2}(k^2 + m^2) |k, \uparrow\rangle \quad (2.55)$$

The first one is zero because of c . From the first of (2.55), we see that $|k, \uparrow\rangle$ is a closed state, i.e, a member of $\ker Q_B$. $|k, \downarrow\rangle$ is closed only when $k^2 + m^2 = 0$. The state $|k, \uparrow\rangle$ is exact (a member of $\text{Im}\{Q_B\}$) if $k^2 + m^2 \neq 0$. Therefore, the closed states that are not exact are,

$$|k, \downarrow\rangle, \quad \text{with} \quad k^2 + m^2 = 0 \quad (2.56)$$

$$|k, \uparrow\rangle, \quad \text{with} \quad k^2 + m^2 = 0 \quad (2.57)$$

Therefore, the physical states must obey the mass shell condition! But we seem to have a problem - there are two identical copies of the same state. This is not something that we know from our previous experience with relativistic point particles. **Polchinski then says that the physical states must satisfy an additional condition that they must be annihilated by b , and gives a kinematic reason for this. So the spectrum is actually a subset of the cohomology. I do not understand this.**

3 The Bosonic String

Let us now study the bosonic string BRST quantization in detail. The Polyakov action is given by,

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} g^{ab} \partial_a X^\mu \partial_b X_\mu \quad (3.1)$$

The action has the 2 dimensional analogue of reparameterization invariance, which is usually referred to as the diffeomorphism (diffeo) invariance,

$$\sigma \rightarrow \sigma'(\sigma) \quad (3.2)$$

Under this change, the metric changes as,

$$g_{ab}(\sigma') = \frac{\partial\sigma^c}{\partial\sigma'^a} \frac{\partial\sigma^d}{\partial\sigma'^b} g_{cd}(\sigma) \quad (3.3)$$

This symmetry is actually manifest in the way the action is written. All the worldsheet indices are contracted. For infinitesimal diffeo,

$$\sigma'^a = \sigma^a - \epsilon \xi^a(\sigma) \quad (3.4)$$

we will have,

$$\delta_E X^\mu(\sigma) = -\epsilon \xi^a(\sigma) \partial_a X^\mu(\sigma) \quad (3.5)$$

and,

$$\delta_E g_{ab}(\sigma) = -\epsilon (\nabla_a \xi_b + \nabla_b \xi_a) \quad (3.6)$$

The subscript E denote “Einstein”.

In addition to this, the string action has one more local symmetry, which is called the Weyl symmetry. This is not the change of any coordinates, but a local re-scaling of the metric,

$$g_{ab}(\sigma) \rightarrow g'_{ab}(\sigma) = e^{2\omega(\sigma)} g_{ab}(\sigma) \quad (3.7)$$

The factor of ω contributed by the square root of the determinant cancels exactly with the inverse metric contribution. The Weyl invariance is rather special to 2 dimensions. Infinitesimally,

$$\delta_W X^\mu(\sigma) = 0 \quad (3.8)$$

and

$$\delta_W g_{ab}(\sigma) = 2\epsilon\omega(\sigma) g_{ab}(\sigma) \quad (3.9)$$

And here the subscript W stands for “Weyl”.

Faddeev-Popov Quantization

We want to quantize string theory via the path integral. The partition function is,

$$Z = \int \mathcal{D}X \mathcal{D}g e^{-S[X,g]} \quad (3.10)$$

where the action is given by (3.1). But, the functional integral over the metric is highly redundant because of the gauge symmetries that we have. It is convenient to denote both diffeo and Weyl as ζ . This means,

$$g \rightarrow g^\zeta \quad \text{where} \quad g_{ab}^\zeta(\sigma') = e^{2\omega(\sigma)} \frac{\partial \sigma^c}{\partial \sigma'^a} \frac{\partial \sigma^d}{\partial \sigma'^b} g_{cd}(\sigma) \quad (3.11)$$

There is a confusing aspect here. What should be the argument of ω ? Is it σ or σ' ? It is clear from (3.7) that Weyl transformation relates the metric at the same spacetime point. But in (3.11), it seems like the transformation relates different points. This is just confusing notation, nothing else. Even for diffeos, the metric transformation should be given at the same point, but the component notation hides it. The actual math notation goes something like $g'(p) = (\Lambda \Lambda g) \circ \phi(p)$, where ϕ is the diffeo. In component notation, we omit writing the ϕ part. Therefore, the argument of ω should be σ , which is the point at which we are comparing the metric before and after gauge transformation.

We would like to fix a gauge. Since we are in 2 dimensions, three symmetries (one Weyl and two diffeos) are enough to fix the metric completely. We hence fix the metric to be equal to $\hat{g}$, which is called the *fiducial metric*. For example, $\hat{g}_{ab} = \delta_{ab}$ will be one very convenient choice that we will use later. We can choose the metric to be $\hat{g}$ by a choice of the gauge parameter ζ_0 . For this purpose, we insert 1 into the partition function in the form,

$$1 = \int \mathcal{D}\zeta \delta(\zeta - \zeta_0) \quad (3.12)$$

We specify the gauge parameter in a different way; we need ζ_0 to be that gauge transformation that takes a given metric g to $\hat{g}$. In other words, we need ζ_0 to be the solution of $g^\zeta = \hat{g}$. This can be done as,

$$1 = \int \mathcal{D}\zeta \delta(\zeta - \zeta_0) = \int \mathcal{D}\zeta \Delta_{FP}[\hat{g}] \delta(g^\zeta - \hat{g}) \quad (3.13)$$

The Jacobian factor is known as the Faddeev-Popov determinant. Let us insert this into the path integral,

$$Z = \int \mathcal{D}\zeta \int \mathcal{D}X \mathcal{D}g \Delta_{FP}[\hat{g}] \delta(g^\zeta - \hat{g}) \exp\{-S[X, g]\} \quad (3.14)$$

Since the action is gauge invariant, and the integration measure $\mathcal{D}g$ is also assumed to be gauge invariant³, can re-write this as,

$$Z = \int \mathcal{D}\zeta \int \mathcal{D}X \mathcal{D}g^\zeta \Delta_{FP}[g^\zeta] \delta(g^\zeta - \hat{g}) \exp\{-S[X, g^\zeta]\} \quad (3.15)$$

Aside : FP determinant is also gauge invariant

We have $\Delta_{FP}[g] = \Delta_{FP}[g^\zeta]$. This can be shown as,

$$(\Delta_{FP}[g^\zeta])^{-1} = \int \mathcal{D}\zeta' \delta(g^{\zeta'} - \hat{g}) = \int \mathcal{D}\zeta'' \delta(g^{\zeta''} - \hat{g}) = (\Delta_{FP}[g])^{-1} \quad (3.16)$$

here we have use the invariance of the measure $\mathcal{D}(\zeta\zeta') = \mathcal{D}\zeta'$ for a fixed ζ .

The metric integral can be done now, which is what usually is done. Since we want to get the full gauge fixed action to figure out BRST, let us not do the integral yet. Let us now just rename $g^\zeta \rightarrow g$, and re-write the delta function as,

$$\delta(g - \hat{g}) = \int \mathcal{D}B \exp\left\{\frac{i}{4\pi} \int d^2\sigma \sqrt{g} B_{ab}(g^{ab} - \hat{g}^{ab})\right\} \quad (3.17)$$

Let us now evaluate the FP determinant. We know the inverse of the FP determinant is,

$$\Delta_{FP}^{-1}[\hat{g}] = \int \mathcal{D}\zeta \delta(g^\zeta - \hat{g}) \quad (3.18)$$

The delta function has support only for one particular ζ . Let us understand the FP determinant locally. If we are considering only infinitesimal ζ , then g should better be close to $\hat{g}$ to start with, otherwise we won't have non zero support for the delta function. Therefore, let us consider $\delta(\hat{g}^\zeta - \hat{g})$ for infinitesimal ζ . Let ω be the infinitesimal Weyl transformation and $\delta\sigma^a \equiv v^a(\sigma)$ be the infinitesimal diffeo. Then,

$$\delta\hat{g}_{ab} = \hat{g}_{ab}^\zeta - \hat{g}_{ab} = 2\omega(\sigma)\hat{g}_{ab} - \hat{\nabla}_a v_b(\sigma) - \hat{\nabla}_b v_a(\sigma) \quad (3.19)$$

The “hat” on the covariant derivative means that the metric involved is $\hat{g}$. Let us define an operator that will take a vector to a traceless symmetric 2 tensor with the help of the covariant derivative as,

$$(\hat{P}v)_{ab} := \frac{1}{2} \left(\hat{\nabla}_a v_b(\sigma) + \hat{\nabla}_b v_a(\sigma) - \hat{g}_{ab} \hat{\nabla}_c v^c(\sigma) \right) \quad (3.20)$$

³the measure is actually not gauge invariant, and hence the reason for Weyl anomaly. But with the ghost action (or FP determinant) in place, we can arrange for matters in such a way that the Weyl anomaly cancels! This arrangement is letting X^μ be a 26 dimensional vector.

Then,

$$\delta\hat{g}_{ab} = (2\omega(\sigma) - \hat{\nabla}_c v^c(\sigma))g_{ab} - 2(\hat{P}v)_{ab} \quad (3.21)$$

Therefore,

$$\begin{aligned} \Delta_{FP}^{-1}[\hat{g}] &= \int \mathcal{D}\omega \mathcal{D}v^a \delta\left(-(2\omega - \hat{\nabla} \cdot v)\hat{g} + 2\hat{P}v\right) \\ &= \int \mathcal{D}\omega \mathcal{D}v^a \int \mathcal{D}\beta^{ab} \exp\left\{2\pi i \int d^2\sigma \sqrt{\hat{g}} \beta^{ab} \left(-(2\omega - \hat{\nabla} \cdot v)\hat{g} + 2\hat{P}v\right)_{ab}\right\} \end{aligned} \quad (3.22)$$

β^{ab} is a symmetric tensor. We can now perform the integral over ω , which sets the trace of β to be zero. Then the functional integral over β changes to that of β' , where β' is a symmetric, traceless 2 tensor. Thus,

$$\Delta_{FP}^{-1}[\hat{g}] = \int \mathcal{D}\beta \mathcal{D}v \exp\left\{4\pi i \int d^2\sigma \sqrt{\hat{g}} \beta'^{ab} (2\hat{P}v)_{ab}\right\} \quad (3.23)$$

But using the symmetry and tracelessness of β' , we can write,

$$\beta'^{ab} (2\hat{P}v)_{ab} = \beta'^{ab} \hat{\nabla}_a v_b \quad (3.24)$$

And, we can turn the inverse determinant to a determinant by replacing the functional integrals with Grassmann odd variables,

$$v^a \rightarrow c^a \quad \text{and} \quad \beta'_{ab} \rightarrow b_{ab} \quad (3.25)$$

We thus get,

$$\Delta_{FP}[\hat{g}] = \int \mathcal{D}b \mathcal{D}c \exp\left\{-\frac{1}{2\pi} \int d^2\sigma \sqrt{\hat{g}} b_{ab} \hat{\nabla}^a c^b\right\} \quad (3.26)$$

Where in the last step, we have re-scaled the fields for suitable normalization. Thus, the full gauge fixed path integral looks like,

$$Z = \int \mathcal{D}X \mathcal{D}g \mathcal{D}B \mathcal{D}b \mathcal{D}c \, e^{-S - S_{\text{gf}} - S_{\text{gh}}} \quad (3.27)$$

where,

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} g^{ab} \partial_a X^\mu \partial_b X_\mu, \quad (3.28)$$

$$S_{\text{gf}} = -\frac{i}{4\pi} \int d^2\sigma \sqrt{g} B_{ab} (g^{ab} - \hat{g}^{ab}), \quad (3.29)$$

and,

$$S_{\text{gh}} = \frac{1}{2\pi} \int d^2\sigma \sqrt{\hat{g}} b_{ab} \hat{\nabla}^a c^b \quad (3.30)$$

BRST

Let us choose $\hat{g}_{ab} = \delta_{ab}$. The Weyl transformation was integrated out already, and is accounted for by the fact that b_{ab} is traceless. The gauge transformations then are,

$$\delta_E X^\mu(\sigma) = -\epsilon v^a(\sigma) \partial_a X^\mu(\sigma) \quad (3.31)$$

This is the two dimensional version of the point particle diffeo, hence the algebra of a basis of gauge transformations is the same. We choose the basis as,

$$\delta_{\sigma_1^a} \sigma^b(\sigma) = -\delta_a^b \delta^2(\sigma - \sigma_1) \quad (3.32)$$

Then the algebra of the basis will have a similar set of structure functions. We can hence construct the BRST transformations by analogy. For X^μ ,

$$\delta_B X^\mu(\sigma) = i\epsilon c^a(\sigma) \partial_a X^\mu(\sigma) \quad (3.33)$$

Also,

$$\delta_B b_{ab} = \epsilon B_{ab} \quad (3.34)$$

We will solve for B_{ab} via the equations of motion of g_{ab} . If we vary g^{ab} in S , we get the energy-momentum tensor. By definition,

$$T_{ab}^m(\sigma) = -\frac{4\pi}{\sqrt{g}} \frac{\delta S}{\delta g^{ab}(\sigma)} \quad (3.35)$$

Varying the ghost action with g^{ab} gives the energy momentum tensor of the ghost system. If we do the same with S_{gf} , (and set $g_{ab} = \delta_{ab}$ on account of integrating out B), we get,

$$\delta_B b_{ab}(\sigma) = i\epsilon(T_{ab}^m(\sigma) + T_{ab}^g(\sigma)) \quad (3.36)$$

c^a transforms as,

$$\delta_B c^a(\sigma) = i\epsilon c^b \partial_b c^a \quad (3.37)$$

We can now switch to complex metric. Denote,

$$c(z) = c^z, \quad \tilde{c}(\bar{z}) = c^{\bar{z}} \quad (3.38)$$

and

$$b(z) = b_{zz} \quad \tilde{b}(\bar{z}) = b_{\bar{z}\bar{z}} \quad (3.39)$$

Then the BRST transformations can be written as,

$$\begin{aligned}
\delta_B X^\mu(z, \bar{z}) &= i\epsilon(c(z)\partial + \tilde{c}(\bar{z})\bar{\partial})X^\mu(z, \bar{z}) \\
\delta_B b(z) &= i\epsilon(T^m(z) + T^g(z)) \\
\delta_B \tilde{b}(\bar{z}) &= i\epsilon(\tilde{T}^m(\bar{z}) + \tilde{T}^g(\bar{z})) \\
\delta_B c(z) &= i\epsilon(c\partial + \tilde{c}\bar{\partial})c \\
\delta_B \tilde{c}(\bar{z}) &= i\epsilon(c\partial + \tilde{c}\bar{\partial})\tilde{c}
\end{aligned} \tag{3.40}$$

The BRST variations of c and $\tilde{c}$ vanish because of the EOM. To see this, in $g_{ab} = \delta_{ab}$ gauge, we have the ghost action to be,

$$S_{\text{gh}} = \frac{1}{2\pi} \int d^2z (b\partial c + \tilde{b}\bar{\partial}\tilde{c}) \tag{3.41}$$

The EOM for b and $\tilde{b}$ are,

$$\partial c(z) = 0 \quad \text{and} \quad \bar{\partial}\tilde{c}(\bar{z}) = 0 \tag{3.42}$$

This along with the fact that c (or $\tilde{c}$) is holomorphic (or anti holomorphic) makes the BRST transformation for c and $\tilde{c}$ vanish.

BRST Current J_B

The total action now is,

$$S = \frac{1}{2\pi\alpha'} \int d^2z \partial X^\mu \bar{\partial} X_\mu + \frac{1}{2\pi} \int d^2z (b\partial c + \tilde{b}\bar{\partial}\tilde{c}) \tag{3.43}$$

4 BRST Quantization of Electromagnetic Field

Faddev-Popov Quantization

The electromagnetic field is described by the Lagrangian,

$$\mathcal{L}_{EM} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} \quad (4.1)$$

Where $F_{\mu\nu}$ is the *Field Strength* tensor, defined in terms of the vector potential A_μ as,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (4.2)$$

The Maxwell Lagrangian (4.1) has a symmetry (on top of the usual $ISO(3,1)$ symmetry), called the *Gauge Symmetry*. It is actually a redundancy, rather than a symmetry. Gauge symmetry states that $A_\mu(x)$ and $A_\mu(x) - \partial_\mu\alpha(x)$ describe the same physics, for any function $\alpha(x)$. We can make the substitution,

$$A'_\mu = A_\mu - \partial_\mu\alpha \quad \text{or} \quad \delta A^\mu = -\partial^\mu\alpha \quad (4.3)$$

in (4.2) and see that it does not change. Hence (4.1) also stays invariant.

We would like to *quantize* the Maxwell theory, via the *path integral* method. We hence set out to calculate the generating functional,

$$Z_0[J^\mu] = \int \mathcal{D}A \exp\left\{i \int_x \left(-\frac{1}{4}F^2 + J^\mu A_\mu\right)\right\} \quad (4.4)$$

Here $J_\mu(x)$ is an arbitrary *source function*. We can substitute (4.2) to (4.1) and get it to the following form,

$$\mathcal{L}_{EM} = -\frac{1}{2}(\partial^\mu A^\nu \partial_\mu A_\nu - \partial^\mu A^\nu \partial_\nu A_\mu) \quad (4.5)$$

Integrating by parts, and ignoring the total derivative term gives us,

$$S_{EM} = \int_x -\frac{1}{2}A^\mu(x)(-\eta_{\mu\nu}\square + \partial_\mu\partial_\nu)A^\nu(x) \quad (4.6)$$

Therefore, what we have to compute is,

$$Z_0[J] = \int \mathcal{D}A \exp\left\{i \int_x \left[-\frac{1}{2}A^\mu(-\eta_{\mu\nu}\square + \partial_\mu\partial_\nu)A^\nu + J^\mu A_\mu\right]\right\} \quad (4.7)$$

Let us try to follow the usual method. We know, for gaussian integrands, we have

$$Z_0[J] = \mathcal{N}e^{iS[A_{cl},J]} \quad (4.8)$$

Where, $\mathcal{N}$ is a normalization constant and, $iS[A, J]$ is the functional in the exponent, with A_{cl} given by,

$$0 = \frac{\delta S[A, J]}{\delta A} \Big|_{A=A_{cl}} = -(-\eta_{\mu\nu}\square + \partial_\mu\partial_\nu)A_{cl}^\nu + J_\mu \quad (4.9)$$

Then A_{cl} can be found by the use of *Green's functions*,

$$A_{cl}^\mu(x) = \int_y G^{\mu\nu}(x-y)J_\nu(y) \quad (4.10)$$

With the Green's function obeying,

$$(-\eta_{\mu\nu}\square + \partial_\mu\partial_\nu)G^{\nu\sigma}(x-y) = \delta_\mu^\sigma\delta^4(x-y) \quad (4.11)$$

We can introduce the Fourier transformations,

$$\delta^4(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot(x-y)} \quad (4.12)$$

and,

$$G^{\mu\nu}(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{ik\cdot(x-y)} \tilde{G}^{\mu\nu}(k) \quad (4.13)$$

Then (4.11) gives,

$$(\eta_{\mu\nu}k^2 - k_\mu k_\nu)\tilde{G}^{\nu\sigma}(k) = \delta_\mu^\sigma \quad (4.14)$$

Let us define,

$$P_{\mu\nu} := \eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \quad (4.15)$$

Then naively,

$$\tilde{G}^{\mu\nu}(k) = \frac{(P^{-1})^{\mu\nu}}{k^2} \quad (4.16)$$

This is easy! But there is one caveat. If the matrix P is not invertible, then we are in trouble! In order to check this, we first compute P^2 and find that⁴,

$$P_\nu^\mu P_\sigma^\nu = P_\sigma^\mu \quad (4.17)$$

What this says is that $P(P-1) = 0$ or the matrix P has only eigen-values either 0 or 1. Now, we find the trace of P ,

$$\text{Tr}\{P\} = P_\mu^\mu = 3 \quad (4.18)$$

Hence, the eigen-values of P must be $\{0, 1, 1, 1\}$. Since one of the eigen-values is zero, the matrix P_σ^ν is not invertible!

⁴Of course, this is left as an exercise!

How do we proceed now? To get a better understanding of the issue, let us find the eigenvector corresponding to the zero eigen-value. This is very simple, and we find that any vector of the form $\tilde{\alpha}(k)k^\mu$ has zero eigen-value, where $\tilde{\alpha}(k)$ is any arbitrary function. In the path integral, we are integrating over all $\tilde{A}_\mu(k)$. If for some reason, we can omit integrating along the direction of k^μ , then we are safe. This would correspond to restricting $\tilde{A}_\mu(k)$ to be in a subspace orthogonal to k_μ . But does this loose any relevant information? Let's find out. In position space,

$$\tilde{\alpha}(k)k^\mu \longrightarrow -i\partial_\mu\alpha(x) \quad (4.19)$$

Gauge symmetry tells us that $A_\mu(x)$ and $A_\mu(x) - \partial_\mu\alpha(x)$ are equivalent field configurations. Hence even if we drop the integral along $\partial_\mu\alpha(x)$, we are not loosing any relevant information! Therefore, we first re-write (4.7),

$$Z_0[J] = \int \mathcal{D}\alpha \mathcal{D}A_{phys} e^{iS[A,J]} \quad (4.20)$$

where A_{phys} are physically distinct field configurations, and the integral over α covers the equivalent field configurations for each A_{phys} . We have just seen that integral over α renders P non-invertible, but this integral over α is completely redundant as far as the physics is concerned! Hence we drop the integral over α , but in a fancy way⁵. We insert a delta functional to the integral,

$$Z_0[J] = \int \mathcal{D}\alpha \mathcal{D}A_{phys} e^{iS[A,J]} \delta(\alpha(x) - \alpha_0(x)) \quad (4.21)$$

This is called *Gauge Fixing*. But we are usually used to fixing the gauge via *gauge conditions*, for example the *Lorenz Gauge*,

$$\partial_\mu A^\mu(x) = 0 \quad (4.22)$$

The way α is chosen is the following: suppose to start with we have $f(x) = \partial_\mu A^\mu(x) \neq 0$. Then we gauge transform to A'_μ such that $\partial_\mu A'^\mu(x) = 0$. Therefore,

$$0 = \partial_\mu A'^\mu(x) = \partial_\mu(A^\mu - \partial^\mu\alpha(x)) = f(x) - \square\alpha(x) \quad (4.23)$$

We just have to choose α such that the above equation is true. One issue is that this does not completely fix the gauge. We can add $\beta(x) \in \ker \square$ to α and this still solves the above equation. Here, I am going to treat this without any rigor. Look up *Gribov Ambiguities* to learn more! We now generalize the identity,

$$\delta^n(\mathbf{f}(\mathbf{x})) = \sum_i \frac{\delta^n(\mathbf{x} - \mathbf{x}_i)}{\left| \det \left\{ \frac{\partial f_i}{\partial x_j} \right\} \right|} \quad (4.24)$$

⁵This is keeping in mind future generalizations to Yang-Mills theory

Where i runs over all solutions of $\mathbf{f}(\mathbf{x}) = \mathbf{0}$. Likewise in the functional case,

$$\delta(G(A)) = \sum_{\beta \in \ker \square} \frac{\delta(\alpha - \alpha_0)}{\det\left\{\frac{\delta G}{\delta \alpha}\right\}} \quad (4.25)$$

We choose,

$$G(A) = \partial_\mu A^\mu(x) - \omega(x) \quad (4.26)$$

where ω is any arbitrary function. This is Lorenz gauge, slightly generalized. G is called the *Gauge Fixing Function*. Notice we omitted the absolute value. This is also an assumption, which is not entirely true. Rectification is done via Gribov ambiguities. I do not know much about it.

We now find the functional derivative. The variation is with respect to gauge transformations. Under gauge transformations,

$$A' = A - \partial\alpha \Rightarrow \delta A = -\partial\alpha \quad (4.27)$$

Hence,

$$\delta G(A) = \partial\delta A - \delta\omega = -\square\alpha \quad (4.28)$$

ω is an arbitrary function and has nothing to do with gauge transformations. Thus,

$$\frac{\delta G}{\delta \alpha} = -\square_x \delta^4(x - y) \quad (4.29)$$

Therefore,

$$\sum_{\beta \in \ker \square} \delta(\alpha - \alpha_0) = \det\{-\square\} \delta(G(A)) \quad (4.30)$$

Let $\text{Vol} = \text{“Volume}(\ker \square)\text{”}$. Then if we do functional integration on both sides,

$$\det\{-\square\} \int \mathcal{D}\alpha \delta(G(A)) = \int \mathcal{D}\alpha \sum_{\beta \in \ker \square} \delta(\alpha - \alpha_0) = \text{Vol} \quad (4.31)$$

Therefore, insert $\frac{1}{\text{Vol}} \delta(\alpha - \alpha_0)$ to (4.7), which gives us,

$$Z_0[J] = \frac{1}{\text{Vol}} \det\{-\square\} \int \mathcal{D}\alpha \mathcal{D}A_{phys} \delta(\partial_\mu A^\mu(x) - \omega(x)) e^{iS[A, J]} \quad (4.32)$$

One way to re-write the determinant is using Grassmann-Odd variables $\bar{c}$ and c ,

$$\det\{-\square_x \delta^4(x - y)\} \propto \int \mathcal{D}c \mathcal{D}\bar{c} e^{i \int_{x,y} \bar{c}(x) (\square_x \delta^4(x-y)) c(y)} = \int \mathcal{D}c \mathcal{D}\bar{c} e^{i \int_x \bar{c}(x) \square c(x)} \quad (4.33)$$

The G-Odd variables have a Lagrangian that is like that of a scalar. For this reason, they are called *Faddeev-Popov Ghosts*. However, in QED the ghosts do not couple to A , hence we

can forget about them. But to extend the gauge symmetry to BRST, we will need them even in QED. Let us omit them for now.

It would be great if we can get rid of the delta function somehow. We notice that the partition function does not depend on the choice of ω . Hence we can average over ω to get rid of the delta function. But just integrating over ω on the left side is meaningless, so we should include a weight factor that is convergent under ω integral. One choice is,

$$\exp\left\{-\frac{i}{2\xi}\int_x\omega^2(x)\right\} \quad (4.34)$$

where $\xi \in \mathbb{R}$ (We will briefly see why other choices would not make sense soon!). Therefore, we take the average over both sides weighted by this gaussian,

$$\int \mathcal{D}\omega e^{-\frac{i}{2\xi}\int_x\omega^2(x)} Z_0[J] = \int \mathcal{D}\omega e^{-\frac{i}{2\xi}\int_x\omega^2(x)} \frac{1}{\text{Vol}} \det\{-\square\} \int \mathcal{D}\alpha \mathcal{D}A_{phys} \delta(\partial_\mu A^\mu(x) - \omega(x)) e^{iS[A,J]} \quad (4.35)$$

On the left side, this will just give a number, while on the right side, we get rid of the delta function. Let us collect all the normalization constants ($\det\{-\square\}$, Vol and the ω integral number from LHS included) to $\mathcal{N}$. Then,

$$Z_0[J] = \mathcal{N} \int \mathcal{D}\alpha \mathcal{D}A_{phys} \exp\left\{i \int_x \left[-\frac{1}{2}A^\mu(-\eta_{\mu\nu}\square + \partial_\mu\partial_\nu)A^\nu + J^\mu A_\mu - \frac{i}{2\xi}(\partial_\mu A^\mu)^2\right]\right\} \quad (4.36)$$

Since the delta function satisfied, the gauge is fixed now. Hence the integral over α is redundant, and will yield only an infinite constant. We absorb this to $\mathcal{N}$. Hence (after integration by parts),

$$Z_0[J] = \int \mathcal{D}A_{phys} \exp\left\{i \int_x \left[-\frac{1}{2}A^\mu\left(-\eta_{\mu\nu}\square + \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right)A^\nu + J^\mu A_\mu\right]\right\} \quad (4.37)$$

Now, we follow the usual method. We first need to solve,

$$\left(-\eta_{\mu\nu}\square + \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right)A^\nu_{cl}(x) = J_\mu(x) \quad (4.38)$$

To find the Green's function, we must invert,

$$P^{\mu\nu} := \eta^{\mu\nu} - \left(1 - \frac{1}{\xi}\right)\frac{k^\mu k^\nu}{k^2} \quad (4.39)$$

This is invertible, and the inverse is⁶

$$(P^{-1})^{\mu\nu} = \eta^{\mu\nu} - (1 - \xi)\frac{k^\mu k^\nu}{k^2} \quad (4.40)$$

⁶try finding this as an exercise, or verify the result yourself!

Notice that the inverse blows up only for $\xi \rightarrow \infty$ which brings us back to the old action. Hence, the Green's function is given by,

$$G^{\mu\nu}(x-y) = \int \frac{d^4k}{(2\pi)^4} e^{ik \cdot (x-y)} \frac{1}{k^2} \left(\eta^{\mu\nu} - (1-\xi) \frac{k^\mu k^\nu}{k^2} \right) \quad (4.41)$$

Each choice of ξ corresponds to different avatars of the Lorenz gauge. $\xi = 1$ is called the Feynman-t'Hooft gauge.

Now we can solve for,

$$S[A_{cl}, J] = \int_x \frac{1}{2} J_\mu(x) A_{cl}^\mu(x) = \frac{1}{2} \int_x \int_y J_\mu(x) G^{\mu\nu}(x-y) J_\nu(y) \quad (4.42)$$

We introduce diagrammatic notation,

$$i \sum_\mu \int_x J^\mu(x) = \text{blob} \quad (4.43)$$

(unmarked blob is a source, and summed over)

$$\frac{1}{i} G^{\mu\nu}(x-y) = \text{wavy line} \quad (4.44)$$

Thus,

$$\frac{1}{2} \int_x \int_y J_\mu(x) G^{\mu\nu}(x-y) J_\nu(y) = \frac{1}{2} \text{blob} \text{---} \text{wavy line} \text{---} \text{blob} \quad (4.45)$$

Now using the normalization $Z_0[J=0] = 1$, we arrive at,

$$Z_0[J] = \exp \left\{ \frac{1}{2} \text{blob} \text{---} \text{wavy line} \text{---} \text{blob} \right\} \quad (4.46)$$

4.1 BRST Symmetry : Take 1

After the Faddev-Popov procedure, we have a *gauge fixed* Lagrangian,

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} - \underbrace{\frac{1}{2\xi} (\partial_\mu A^\mu)^2}_{\mathcal{L}_{gf}} - \underbrace{\partial^\mu \bar{c} \partial_\mu c}_{\mathcal{L}_{gh}} \quad (4.47)$$

This Lagrangian do not have gauge symmetry, precisely because we fixed a gauge. But it turns out that the Lagrangian still have a remaining symmetry, called the BRST⁷ symmetry.

⁷Named after Carlo **B**ecchi, Alain **R**ouet, Raymond **S**tora and Igor **T**yutin

Roughly speaking, we derive BRST symmetry by upgrading the gauge parameter to be the ghost c . Let us first re-write the Lagrangian in terms of an auxiliary field $B(x)$ as,

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - B(x)\partial_\mu A^\mu + \frac{1}{2}\xi B^2(x) - \partial^\mu \bar{c}\partial_\mu c \quad (4.48)$$

It is quite easy to see that solving the EOM for B and plugging back to (4.48) will give back (4.47). Let,

$$\delta_B A^\mu(x) = \epsilon \partial^\mu c(x) \quad (4.49)$$

Notice that this is almost the gauge variation (4.3), but the gauge parameter α is replaced by the ghost $-\epsilon c$. $\epsilon \in \mathbb{R}$ is the infinitesimal parameter, and most importantly is *independent of* x . Since the right hand side is Grassmann odd, we must have δ_B a Grassmann odd variation to have a sensible equation⁸. We now demand (4.49) is a symmetry of the action. Or in other words,

$$\delta_B S = 0 \quad (4.50)$$

We know that the first term F^2 in (4.48) is already invariant under (4.49). The argument is the same as what we made for gauge transformation (4.3). Hence, we must have,

$$\delta_B \left(-B(x)\partial_\mu A^\mu + \frac{1}{2}\xi B^2(x) - \partial^\mu \bar{c}\partial_\mu c \right) = \partial_\nu K^\nu \quad (4.51)$$

where K^ν is some arbitrary function (Lagrangian can vary upto a total derivative). Modulo a total derivative, we must have,

$$-(\delta_B B)\partial_\mu A^\mu - \epsilon B \square c + \xi B \delta_B B - \partial^\mu (\delta_B \bar{c})\partial_\mu c + \partial^\mu \bar{c}\partial_\mu (\delta_B c) = 0 \quad (4.52)$$

Where the last term got a minus sign because of the anticommuting nature of the variation δ_B . We can re-write these as,

$$-(\delta_B B)\partial_\mu A^\mu - \epsilon B \square c + \xi B \delta_B B + (\delta_B \bar{c})\square c - \underbrace{\partial^\mu (\delta_B \bar{c}\partial_\mu c)}_{\text{we can drop}} + \partial^\mu \bar{c}\partial_\mu (\delta_B c) = 0 \quad (4.53)$$

We can drop the total derivative. Staring at the above equation for a while, we see that the following recipe will satisfy BRST invariance,

$$\begin{aligned} \delta_B A^\mu(x) &= \epsilon \partial^\mu c(x) \\ \delta_B c(x) &= 0 \\ \delta_B \bar{c}(x) &= \epsilon B(x) \\ \delta_B B(x) &= 0 \end{aligned} \quad (4.54)$$

⁸We already anticipate $\delta_B^2 = 0$

These transformations are called the BRST transformations⁹. The BRST transformations are *NOT* gauge transformations. Under gauge transformations, only A change. Every other field stay intact. Since the BRST parameter ϵ is independent of x , BRST is actually a symmetry rather than a redundancy!(Do not confuse c to be the BRST parameter by drawing analogy with the gauge parameter $\alpha(x)$. We call α the gauge parameter because it never made an appearance in the Lagrangian. It was not a field. Here, c is part of the Lagrangian (i.e, it is a field), but ϵ , which we now call the BRST parameter is not a field, and hence not part of the Lagrangian

Nilpotency of δ_B

If we take (4.54), and try finding,

$$\delta_B^2(\text{any operator composed of } A, B, \bar{c}, c) = 0 \quad (4.55)$$

In short, we have,

$$\delta_B^2 = 0 \quad (4.56)$$

The BRST Charge, Q_B

Since BRST is a global symmetry, not a redundancy, we must have a Noether current associated with it. This is found using,

$$J_B^\mu(x) = \sum_I \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi_I(x))} \delta_B \Phi_I(x) = \epsilon \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu(x))} \partial_\nu c(x) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{c}(x))} B(x) \right) \quad (4.57)$$

Note to self : Find this explicitly.

We then have the BRST charge defined by,

$$Q_B = \int d^3\mathbf{x} J_B^0(x) \quad (4.58)$$

Then, one can re-write BRST transformations using the Generator Q_B ¹⁰,

$$\begin{aligned} \delta_B A^\mu(x) &= i\epsilon[Q_B, A^\mu(x)] = \epsilon \partial^\mu c(x) \\ \delta_B c(x) &= i\epsilon\{Q_B, c(x)\} = 0 \\ \delta_B \bar{c}(x) &= i\epsilon\{Q_B, \bar{c}(x)\} = \epsilon B(x) \\ \delta_B B(x) &= i\epsilon[Q_B, B(x)] = 0 \end{aligned} \quad (4.59)$$

⁹in Yang-Mills, the transformations will be more complicated

¹⁰ Q_B is Grassmann odd

If you are confused about how we got the above equations, then just think about the Heisenberg equations. There, the Hamiltonian is the generator of time translations and δt is the parameter. We can then write the time translations as, $\delta A = i[H, A]\delta t$, which is nothing but the Heisenberg equations. Here, it is exactly the same, but Q_B is the generator and ϵ is the parameter. The use of (anti)-commutation is depending upon the field is (fermion) boson. It is clear that,

$$Q_B^2 = 0 \quad (4.60)$$

The above is an operator equation, meaning $Q_B^2 |\Psi\rangle = 0, \forall |\Psi\rangle \in \mathcal{F}$, the Fock space¹¹¹². Let us define a couple of subsets of $\mathcal{F}$,

$$\text{Im}\{Q_B\} := \left\{ |\Psi\rangle \in \mathcal{F} \mid \exists |\Phi\rangle \in \mathcal{F} : |\Psi\rangle = Q_B |\Phi\rangle \right\} \quad (4.61)$$

In plain English, what this says is that any vector in the *image* of Q_B can be written as Q_B acting on some other vector. We can also define,

$$\ker \{Q_B\} := \left\{ |\Psi\rangle \in \mathcal{F} \mid Q_B |\Psi\rangle = 0 \right\} \quad (4.62)$$

These are vectors which are annihilated by Q_B , called the *kernal* of Q_B . Since $Q_B^2 = 0$, we have an additional relation,

$$\text{Im}\{Q_B\} \subset \ker \{Q_B\} \quad (4.63)$$

We notice that,

$$|\Psi\rangle \in \text{Im}\{Q_B\} \Rightarrow \langle \Psi | \Psi \rangle = 0 \quad (4.64)$$

Therefore, there are non zero vectors in the image which are not normalizable. We want all our physical states to be normalizable. Clearly, we cannot choose the entire Fock space $\mathcal{F}$ to be our physical set of states. Looks like $\ker \{Q_B\}$ is a good candidate for $\mathcal{H}$. States in the kernal will trivially satisfy the nilpotency condition for Q_B . But since $\text{Im}\{Q_B\} \subset \ker \{Q_B\}$, and vectors in $\text{Im}\{Q_B\}$ have zero norm, we cannot choose the physical Hilbert space to be $\ker \{Q_B\}$ also. One possiblity is to just take away the image from the kernal. This will result in taking away the zero element from the kernal, then we cannot make the kernal into a Hilbert space. We also notice that if $|\Psi\rangle \in \ker \{Q_B\}$ and has a non zero norm, and $|\Phi\rangle \in \text{Im}\{Q_B\}$, then $|\Psi\rangle + |\Phi\rangle \in \ker \{Q_B\}$ and $|\Psi\rangle + |\Phi\rangle$ is normalizable. That is, we can add a vector from the image to the physical state and nothing changes. This gives us an idea: first, define an equivalence relation,

$$|\Psi\rangle \sim |\Phi\rangle \quad \text{if} \quad |\Psi\rangle - |\Phi\rangle \in \text{Im}\{Q_B\} \quad (4.65)$$

¹¹this is the entire Fock space, not the physical set of states

¹²Also, the eigen-values of Q_B hence have to be Grassman-valued.

Then we define a new set called the *Cohomology* of Q_B as,

$$\text{Coh}\{Q_B\} := \ker \{Q_B\} / \text{Im}\{Q_B\} \quad (4.66)$$

The cohomology is the *Quotient* of the kernel with respect to the equivalence class. In plain English, if two elements in the kernel only differ by a vector from the image, then these two elements in the kernel are identified and is one element in the cohomology. This way, all vectors in $\text{Im}\{Q_B\}$ are the same element in the cohomology, namely the zero element. Hence, we define the physical set of states,

$$\mathcal{H} = \text{Coh}\{Q_B\} \quad (4.67)$$

Let us verify that once we start with $|\Psi\rangle \in \text{Coh}\{Q_B\}$, we stay in $\text{Coh}\{Q_B\}$ as time progress. This is easy. If $Q_B |\Psi\rangle = 0$, and $\langle\Psi|\Psi\rangle = 1$ say, then,

$$|\Psi(t)\rangle = e^{-iHt} |\Psi(0)\rangle \quad (4.68)$$

Then since $[Q_B, H] = 0$,

$$Q_B |\Psi(t)\rangle = Q_B e^{-iHt} |\Psi(0)\rangle = e^{-iHt} Q_B |\Psi(0)\rangle = 0 \quad (4.69)$$

Therefore, time evolution keeps the state in the cohomology.

4.2 BRST Symmetry : Take 2

This is a different point of view of BRST. We start of with the usual Maxwell action,

$$S_{EM} = \int_x -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \quad (4.70)$$

We know that the action has local gauge redundancy. Now, we *demand* that the action should have a global BRST symmetry, under which the vector potential changes as,

$$\delta_B A^\mu(x) = \epsilon \partial^\mu c(x) \quad (4.71)$$

where c is a Grassmann odd field. We also insist that the BRST symmetry is such that,

$$\delta_B^2 = 0 \quad (4.72)$$

This simply yields,

$$0 = \delta_B^2 A^\mu(x) = \epsilon \partial^\mu (\delta_B c(x)) \Rightarrow \delta_B c(x) = 0 \quad (4.73)$$

Now, we make the claim that BRST is a fundamental symmetry. We can thus add to the Lagrangian a term of the form,

$$\delta_B \mathcal{O} \quad (4.74)$$

and still have the same theory since BRST is nilpotent. To construct some $\mathcal{O}$, the only fields we have at disposal are A and c . We must have some other $\bar{c}$ in order to write a quadratic term for the Grassmann field. Since c is a single component Grassmann field, a Lorentz invariant Dirac Lagrangian for c is not possible¹³. What would be the BRST variation of $\bar{c}$? Let us write,

$$\delta_B \bar{c}(x) = \epsilon B(x) \quad (4.75)$$

Here, $B(x)$ is just a place holder. B will ultimately be related to A, c and $\bar{c}$. Because of the nilpotency, it is clear that,

$$\delta_B B(x) = 0 \quad (4.76)$$

What we choose for $\mathcal{O}$ is completely up to us. One particular choice that we will look at is,

$$\mathcal{O}(x) = \frac{1}{\epsilon} \bar{c}(x) \left[\frac{1}{2} \xi B(x) - \partial^\mu A_\mu(x) \right] \quad (4.77)$$

Then,

$$\delta_B \mathcal{O} = \frac{1}{2} \xi B^2(x) - B(x) (\partial^\mu A_\mu(x)) + \bar{c} \square c \quad (4.78)$$

The sign flip is again due to the Grassmann odd nature of δ_B . Integrating by parts and throwing away the total derivative,

$$\delta_B \mathcal{O} = \mathcal{L}_{gf} + \mathcal{L}_{gh} \quad (4.79)$$

Hence, the new total Lagrangian, which is BRST invariant, but NOT gauge invariant. This is no problem, because we demanded that BRST symmetry is the fundamental principle, not gauge symmetry. This way, BRST justifies the ad-hoc procedure we did in the Faddeev-Popov method!

4.3 Spectrum of the Theory

One might wonder that since we have introduced ghosts in the Lagrangian, aren't they going to appear in the spectrum of the theory? This would be bad news because ghosts violate the

¹³That is, $\bar{c}\not\partial c$ doesn't make sense, because c and $\bar{c}$ are single component

spin statistics theorem¹⁴. But as we will see now, defining the states to be the cohomology saves the day. Let us write down the fields in their mode expansion,

$$A^\mu(x) = \sum_\lambda \int \widetilde{dk} \left(\varepsilon_\lambda^{\mu*}(k) a_\lambda(k) e^{ikx} + \varepsilon_\lambda^\mu(k) a_\lambda^\dagger(k) e^{-ikx} \right) \quad (4.80)$$

Here, λ runs over four values.

$$c(x) = \int \widetilde{dk} (c(k) e^{ikx} + c^\dagger(k) e^{-ikx}) \quad (4.81)$$

$$\bar{c}(x) = \int \widetilde{dk} (\bar{c}(k) e^{ikx} + \bar{c}^\dagger(k) e^{-ikx}) \quad (4.82)$$

Remember, in these expressions, the k in the exponents are on-shell. Also since both the photon and the ghosts are massless, we have $k^2 = 0$. For simplicity, let us choose,

$$k = \omega(1, 0, 0, 1) \quad (4.83)$$

Then we use the first of (4.59), to get,

$$\sum_\lambda \int \widetilde{dk} \left(\varepsilon_\lambda^{\mu*}(k) [Q_B, a_\lambda(k)] e^{ikx} + \varepsilon_\lambda^\mu(k) [Q_B, a_\lambda^\dagger(k)] e^{-ikx} \right) = \int \widetilde{dk} (c(k) k^\mu e^{ikx} - c^\dagger(k) k^\mu e^{-ikx}) \quad (4.84)$$

This gives us,

$$\sum_\lambda \varepsilon_\lambda^\mu(k) [Q_B, a_\lambda^\dagger(k)] = -k^\mu c^\dagger(k) \quad (4.85)$$

Using $\varepsilon_\lambda^\mu \varepsilon_{\lambda'}^\nu \eta_{\mu\nu} = \delta_{\lambda\lambda'}$, we get,

$$[Q_B, a_\lambda^\dagger(k)] = -\varepsilon_\lambda^\mu k_\mu c^\dagger(k) \quad (4.86)$$

Let us choose the polarization vectors to be,

$$\varepsilon_\lambda^\mu = \delta_\lambda^\mu \quad (4.87)$$

(This is a slightly misleading notation. The λ is not a Lorentz index)

Then (4.86) tells us that $[Q_B, a_{1,2}^\dagger] = 0$ while the other two commutators are not zero (in general, the creation operators which creates polarizations transverse to k^μ are commuting with Q_B). What this means is that if we start with $|\Psi\rangle \in \text{Coh}\{Q_B\}$, and we create a state with polarizations which are not transverse to k_μ , then,

$$Q_B a_{0,3}^\dagger |\Psi\rangle \neq 0 \quad (\text{in more generality, } Q_B a_\parallel^\dagger |\Psi\rangle \neq 0) \quad (4.88)$$

¹⁴essentially, ghosts have a Lagrangian that of a complex scalar, yet the quatization must be done using anti-commutators. I do not know full details of spin statistics theorem. **Note to self : Read up on it more!**

Hence $a_{\parallel}^{\dagger} |\Psi\rangle \notin \text{Coh}\{Q_B\}$, which means they are not physical! Also, if we create a ghost state, $c^{\dagger} |\Psi\rangle$, then (4.86) tells us that,

$$c^{\dagger} |\Psi\rangle \propto [Q_B, a_{\parallel}^{\dagger}] |\Psi\rangle = Q_B a_{\parallel}^{\dagger} |\Psi\rangle \in \text{Im}\{Q_B\} \quad (4.89)$$

Hence $c^{\dagger} |\Psi\rangle$ are also not physical! The same way (by looking at the other non-zero result from (4.59)) we can show that the anti ghosts also are not physical¹⁵. In short, the conclusion is : The ghosts themselves are not physical, and they render the longitudinally polarized modes of the photons also unphysical!

¹⁵Try it!